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Valuing the Benefits of Health and Safety Regulation[1]

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A goal of many public regulatory programs is the protection or enhancement of human health. If such programs are to be subjected to close analytical scrutiny, if not full-blown cost–benefit analysis, it is important to have correct measures of their benefits. This is no simple task. Not only is it often difficult to map regulatory policy into physical improvements in health or safety (less sickness, fewer accidents, etc.), but also it is difficult to translate changes in health status into dollar valuations. This is perhaps most difficult when the policy outputs are "lives saved," or—more correctly—lives prolonged, but difficulties arise even when health benefits take the form of reductions in acute or chronic illness unrelated to premature mortality.

Perhaps the most common means of valuing reduced morbidity is the cost of illness (COI) approach, whereby benefits are valued by savings in direct, out-of-pocket expenses arising from illness or injury (e.g., medicines and doctor and hospital charges), as well as opportunity costs incurred, foregone earnings being the most obvious example (Cooper and Rice [2]; Lave and Seskin [9]).[2] One problem with the COI approach is that it ignores any disutility arising from illness or injury. Perhaps more importantly, the COI approach generally does not take into account the defensive or averting expenditures that individuals can and do make to protect their well-being (Mishan [10]; Fisher and Zeckhauser [5]; and, more recently, Courant and Porter [3]; Shibata and Winrich [12]; Harford [6]). This omission can be critical. If defensive or averting measures fully protect

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[2]As is often the case with ad hoc constructs like the cost of illness, not everyone uses the same definition. Most studies of particular illness incidence, including pollution studies, count only medical care and lost wages. However, attempts to estimate total health care costs sometimes include what we have termed "defensive expenditures" as a cost of illness. See Hodgson and Meiners [7] for a discussion.

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against insults to safety and health, a regulation-induced reduction in such insults might not lead to any observed health improvement, but would be beneficial nonetheless by alleviating the need for such protective measures.

Recently, Courant and Porter [3] and Harford [6] have looked at changes in averting expenditures alone (in a nonhealth framework) as a possible measure of willingness to pay for the alleviation of insults. They conclude that averting expenditures may under- or overstate true benefits. This raises an important question for practical health benefit estimation: If data were available for both the direct and the indirect costs of illness or injury as well as the averting measures, what would be the relationship between their sum and true willingness to pay? This is the question that we address below.

We first present a simple static model of constrained utility maximization in which illness or injury not only reduces the time which can be devoted to labor or leisure and induces remedial medical expenditures, but also causes disutility. In this model, individuals can make defensive or averting expenditures to lessen their chances of illness or injury, thus influencing directly their health status. Next, we recast the problem in terms of the indirect utility function, derive the conceptually correct measure of the benefits of reductions in the health threats that individuals face, and compare this measure with the cost of illness plus the change in defensive expenditures. True benefits are found to exceed this sum for all practical cases. In Section 3 we find that this conclusion holds under a number of extensions to the simple model.

1. THE BASIC MODEL

Consider an individual with a utility function of the form

$$U = U(X, L, S) \tag{1}$$

where X represents expenditures on all non-health-related goods,[3] L is leisure time per period, and S is time spent either ill or injured per period ($U_X, U_L > 0$; $U_S < 0$; $U_{XX}, U_{LL}, U_{SS} < 0$). Time spent sick depends on some broadly defined environmental threat which could take the form of air pollution, exposure to a harmful substance in the workplace or in a foodstuff or other consumer product, or other hazard.[4] For simplicity we refer to all such threats as pollution, P.

Sickness also depends on preventive defensive or averting expenditures, D, which can reduce the likelihood of any illness or injury or shorten the

[3] We find it more convenient to make utility a function of expenditures on various classes of goods rather than a composite good, as is customary.

[4] Although the disutility of sickness presumably depends on its intensity as well as its duration, we assume that all illnesses are equally intense.

duration of those that may occur. For instance, if P is interpreted as air pollution, D might take the form of automobile or home air conditioning or air filtration. If P is a drinking water contaminant, D might represent the purchase of bottled water or a filtration system for the tap. If P represents the threat of accident posed by a particularly dangerous stretch of road, D might even take the form of more cautious driving with its attendant time costs.

Thus,

$$S = S(D, P) \tag{2}$$

where $S_D < 0 < S_P$, and $S_{DD}, S_{PP} > 0$. Medical expenses, M, follow any illness or injury and are assumed to be equal to some daily cost, m, times the number of days spent ill or injured. In other words,

$$M = mS(D, P). \tag{3}$$

Finally, the individual is assumed to receive per-period nonwage income I (from capital or transfer payments) and to work for wage rate w.

The individual's decision problem can then be expressed as

$$\max_{X, L, D} \mathscr{L} = U(X, L, S) + \lambda[I + wT - wL - wS(D, P) \\ - X - D - mS(D, P)] + \mu[T - L - S(D, P)] \tag{4}$$

where T is total time available per period. The bracketed expression multiplying λ is the "full income" constraint as introduced by Becker [1]. The second constraint ensures that the sum of leisure time and time spent ill does not exceed the total time available. The first-order conditions are

$$\mathscr{L}_X = U_X - \lambda = 0 \tag{5}$$

$$\mathscr{L}_L = U_L - w\lambda - \mu = 0 \tag{6}$$

$$\mathscr{L}_D = U_S S_D - w\lambda S_D - \lambda - \lambda m S_D - \mu S_D = 0 \tag{7}$$

$$\mathscr{L}_\lambda = I + wT - wL - wS(D, P) - X - D - mS = 0 \tag{8}$$

$$\mathscr{L}_\mu = T - L - S \geq 0; \ \mu(T - L - S) = 0. \tag{9}$$

For the balance of this paper, we assume that all individuals work some positive number of hours per period. Thus, from (9), $\mu = 0$. It follows, then, that (5) and (6) represent the tradeoff between labor and leisure (X is total expenditures on other goods, so that its price is unity).

Equation (7) can be rewritten to shed light on defensive expenditures by individuals. Specifically,

$$\frac{U_S S_D}{\lambda} - wS_D - mS_D = 1. \tag{10}$$

Thus, at the margin, a dollar's worth of defensive expenditure is comprised of three parts: the dollar value of disutility arising from additional sick time ($U_S S_D/\lambda$), the opportunity cost of illness valued at the wage ($-wS_D$), and the out-of-pocket expenses caused by increased illness ($-mS_D$).

To value the health benefits arising from a reduction in P, we employ the indirect utility function (see Courant and Porter [3]; Just *et al.* [8]). The optimal values for the choice variables X, L, and D depend on w, I, and P. The indirect utility function, V, gives the maximum utility obtainable for a given set of parameter values:

$$\begin{aligned} V &= V(I, P, w) \\ &= U(X, L, S(D, P)) \\ &\quad + \lambda[I + wT - wL - wS(D, P) - X - D - mS(D, P)] \\ &\quad + \mu[T - L - S(D, P)] \end{aligned} \tag{11}$$

where X, L, D, λ, and μ are functions of the parameters I, P, and w.

Suppose that I and w are fixed. If P changes, say from P_0 to P_1, utility changes from V_0 to V_1, where

$$V_0 = V(I, P_0, w) \qquad \text{and} \qquad V_1 = V(I, P_1, w). \tag{12}$$

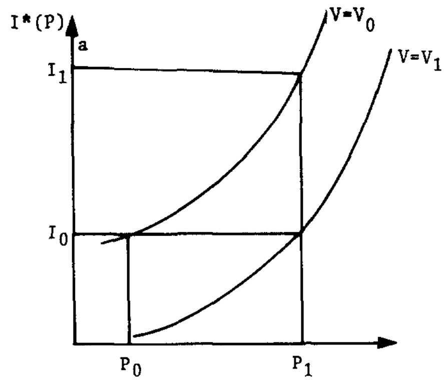
If $P_0 < P_1$, then $V_0 > V_1$. For fixed wages, the only way to keep an individual indifferent to a change in pollution is to change I. This is illustrated in Fig. 1a. When P increases from P_0 to P_1 with no change in income, the individual drops from the indifference curve $V = V_0$ to $V = V_1$. To leave the individual indifferent to the change in P, I must be increased from I_0 to I_1. We define the benefit to the individual of the change in air quality as the measure $I_1 - I_0$.[5]

As suggested by Fig. 1, holding V constant while P varies defines I implicitly as a function of P, which we denote as $I^*(P)$. Thus, $V(I^*(P), P) = V_0$.[6] The total derivative of V with respect to P is set equal to zero along the indifference curve $V = V_0$. Then

$$\frac{dV}{dP} = V_I \frac{dI^*(P)}{dP} + V_P = 0 \qquad \text{or} \qquad \frac{dI^*(P)}{dP} = -\frac{V_P}{V_I}. \tag{13}$$

[5] Note that it is not always possible to keep the individual on the same indifference curve. In the case illustrated in Fig. 1b, there is no change in income that compensates the individual for an increase in pollution from P_0 to P_1. This possibility can arise only for increases in pollution. The change in income associated with a decrease in pollution is always defined. In any event, we are interested primarily in small changes for which this possibility is unlikely to arise; hence, it is ignored in what follows.

[6] Definition of I as a function of P requires $V_I \neq 0$, but we assume, as is customary, that $V_I > 0$.



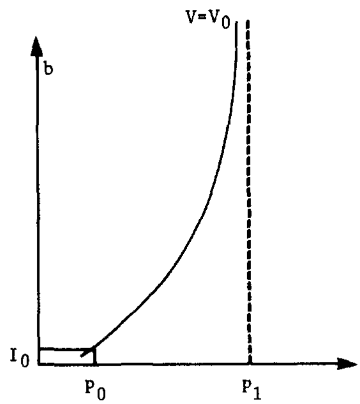


FIG. 1. (a) Indifference curves showing that willingness to pay to avoid increase in pollution from P_0 to P_1 is $I_1 - I_0$. (b) Indifference curve showing that individual cannot be compensated for increase in pollution.

$dI^*(P)/dP$ is the marginal willingness to pay (be compensated) for a marginal decrease (increase) in P. Because V is set equal to V_0, the initial level of utility, in (12), (13) gives the compensating variation. Equivalent variation would be obtained had V been set equal to V_1.

The derivatives of V are easy to compute; because the first-order conditions (5)–(9) hold, many terms drop out. In fact,

$$V_I = \lambda \tag{14}$$

and

$$V_P = (U_S - \lambda w - \lambda m)S_P. \tag{15}$$

But from (7), $U_S - \lambda w - \lambda m = \lambda / S_D$, so that

$$V_P = \lambda \frac{S_P}{S_D} \tag{16}$$

and

$$\frac{dI^*(P)}{dP} = -\frac{V_P}{V_I} = -\frac{S_P}{S_D}. \tag{17}$$

At first glance it would appear that marginal willingness to pay in (17) is independent of personal tastes and income and depends only on the technical characteristics of the sickness (or dose–response) function, S.[7] Note, however, that since D is a choice variable for the individual, the ratio S_P/S_D depends on the characteristics of the utility function and the values of the parameters that the individual faces. S_P/S_D would be independent of D only if $\partial(S_P/S_D)/\partial D = 0$. This condition would be met if S were linear, for example, but is not generally satisfied.

2. TRUE BENEFITS, THE COST-OF-ILLNESS APPROACH, AND DEFENSIVE EXPENDITURES

Let us now examine the relationship between the measure of benefits given in (17) and that provided by the cost-of-illness plus averting or defensive measures. Typically under the COI approach, a dose–response relationship is estimated in a cross-section epidemiological study, like those of Lave and Seskin [9], Crocker [4], or Ostro [11]. This relationship is then used to predict the effect on health status of a change in P. Finally, increases or decreases in S are valued at w to reflect the opportunity cost of illness, and M is linked to sickness through a function like that in (3).

However, virtually no cross-section epidemiological study has included data on defensive expenditures. Since individuals do take such measures to mitigate or prevent the effects of P, what is actually observed in cross-section studies is the total rather than the partial effect of pollution on health. Assuming that the determinants of health other than D and P are invariant with respect to P, the dose–response relationship $S = S(D, P)$ can be totally differentiated to obtain

$$\frac{dS}{dP} = S_D D_P + S_P. \tag{18}$$

Since this total change in health status for a given change in pollution or other insult is used in the COI approach, the added observable cost of

[7] Courant and Porter [3] derive an expression similar to (17) and appear to misinterpret it in a way described in the text.

illness for a change in pollution can be expressed as

$$\frac{dC}{dP} = w\frac{dS}{dP} + m\frac{dS}{dP}, \tag{19}$$

the terms on the right-hand side denoting the lost wages or leisure time due to pollution-related illness, and the resulting out-of-pocket medical costs, respectively.

But from (7),

$$w + m = \frac{U_S}{\lambda} - \frac{1}{S_D}. \tag{20}$$

Combining (18), (19), and (20), we have

$$\frac{dC}{dP} = (w + m)\frac{dS}{dP} = \left[\frac{U_S}{\lambda} - \frac{1}{S_D}\right]\frac{dS}{dP} = \frac{U_S}{\lambda}\frac{dS}{dP} - D_P - \frac{S_P}{S_D}. \tag{21}$$

Hence,

$$\frac{dI^*}{dP} = -\frac{S_P}{S_D} = w\frac{dS}{dP} + m\frac{dS}{dP} - \frac{U_S}{\lambda}\frac{dS}{dP} + D_P. \tag{22}$$

The true willingness to pay to avoid an increase in pollution is, therefore, the amount resulting from the COI approach [the first two terms on the right-hand side of (22)], plus the last two terms. The term $-(U_S/\lambda)(dS/dP)$ is the dollar value of the disutility of pollution-induced illness. D_P is the change in defensive expenditures associated with an increase in pollution.

If $dS/dP > 0$, then marginal willingness to pay always exceeds the sum of changes in defensive expenditures and the cost of illness. Only if this derivative is negative is the sum of the cost of illness and defensive expenditure an overestimate. It is interesting to compare this result with one obtained by Courant and Porter [3], who investigated the suitability of defensive expenditures alone as a measure of the benefits of environmental improvements. They found that defensive expenditures over- or underestimate WTP depending on the shape of the dose–response function, although an underestimate would be the most likely outcome. We agree. According to (22) the only way that defensive expenditures can exceed WTP is for dS/dP to be negative. In other words, individuals would have to respond to an increase in pollution by increasing defensive expenditures so that health improves. This is the only way that the sum of the cost of illness and defensive expenditures can overestimate WTP. Likewise, the only way the cost of illness can overestimate WTP is for D_P to be negative—less than $(U_S/\lambda)(dS/dP)$, in fact. Although nothing in the model prevents either of these outcomes, both would be extremely unlikely in practice.

3. EXTENSIONS OF THE BASIC MODEL

In this section we examine three extensions of the basic model and discuss their effects on benefit estimation. These extensions are as follows:

(i) P is assumed to enter the utility function directly.

(ii) Sickness is assumed to affect the wage rate.

(iii) Individuals are assumed to receive paid sick leave for a fixed work period.

(i) If P has a direct effect on utility, we write the utility function as $U = U(X, L, S, P)$. The first-order conditions (5)–(9) remain unchanged, but the derivative V_P of the indirect utility function becomes

$$V_P = U_S S_P + U_P - \lambda w S_P - \lambda m S_P = \frac{S_P}{S_D} + U_P. \tag{23}$$

If we work through the arithmetic as in the preceding section, we find that

$$\frac{dI^*}{dP} = -\frac{S_P}{S_D} - \frac{U_P}{\lambda} = (w + m)\frac{dS}{dP} + D_P - \frac{U_S}{\lambda}\frac{dS}{dP} - \frac{U_P}{\lambda}. \tag{24}$$

That is, (dI^*/dP) is the same as in our basic case except for the term $-U_P/\lambda$. In this model, the expression for true benefits contains a derivative of U, which is unobservable. Therefore, even if one had data on defensive expenditures, it would not be possible to estimate benefits precisely. Nonetheless, since $U_P < 0$ (no one likes pollution), allowing P to enter the utility function directly makes it even more farfetched for the sum of the cost of illness plus defensive expenditures to exceed willingness to pay. Again if $dS/dP > 0$, then their sum must be an underestimate.

(ii) As Crocker [4] and others have pointed out, it is plausible that the duration of an individual's illness could affect the wage that he or she faces. Chronically ill individuals might be excluded from some physically demanding but high-paying jobs, for example. Or, individuals might be paid a lower wage because of their frequent absences from work.

Suppose this phenomenon is modeled by allowing w to depend on sickness, i.e., $w = w(S)$ where $w'(S) < 0$.[8] This introduces an unexpected twist. The cost of illness can now be written as $C = w(S)S + mS$ which, when differentiated with respect to P, gives

$$dC/dP = [w(S) + Sw'(S) + m]\,dS/dP. \tag{25}$$

Since $w'(S) < 0$, (25) seems to suggest a smaller cost-of-pollution-related

[8]Although it suffices in this simple model to make the wage rate dependent on current illness, a more dynamic formulation in which the wage depends on the worker's cumulative health history would be more realistic.

illness than if the wage rate were independent of pollution—clearly counterintuitive. Forestalling discussion of this point temporarily, and proceeding as in (18)–(22) above, it can be shown that

$$\frac{dC}{dP} = \frac{dS}{dP}\left[\frac{Us}{\lambda} + w'(S)(T-L) - \frac{1}{S_D}\right] \tag{26}$$

which, making use of (18), can be expressed as

$$\frac{dC}{dP} = \frac{U_S}{\lambda}\frac{dS}{dP} - D_P - \frac{S_P}{S_D} + w'(S)(T-L)\frac{dS}{dP}. \tag{27}$$

But since $(dI^*/DP) = (-S_P/S_D)$,

$$\frac{dI^*}{DP} = \frac{dC}{DP} + D_P - \frac{U_S}{\lambda}\frac{dS}{dP} - w'(S)(T-L)\frac{dS}{dP}. \tag{28}$$

Finally, substituting (25) into (28) and rearranging, we have

$$\frac{dI^*}{dP} = [w(S) - (T-L-S)w'(S) + m]\frac{dS}{dP} + D_P - \frac{U_S}{\lambda}\frac{dS}{dP}. \tag{29}$$

Now we can resolve the apparent contradiction noted in (25). In fact, (25) is not the change in the cost of illness induced by pollution because it overlooks the fact that the pollution increase exerts a cost for every hour worked, not just the hours that one is sick. A more appropriate expression for cost of illness is the bracketed term in (29). Since $T - L - S = W$, or hours worked, this term now includes the loss of wages. The other two terms are, as before, defensive expenditures and direct disutility of illness.

(iii) Many workers lose no pay when ill because of paid sick leave. In this case, lost time due to illness still represents a cost to the economy, but one borne by employers. One can therefore argue that paid sick leave does not affect the total economic benefits of pollution control, but only their distribution.

This argument assumes implicitly that the availability of paid sick leave will not affect the efforts of individuals to avoid sickness.[9] To drop this assumption, we must also drop the full-income constraint, which permits leisure and income to be traded off without restriction. Instead, suppose an individual receives a wage w per hour for W hours of work per year, regardless of whether those hours are actually worked. Let $Y = wW$ denote income, and suppose further that the distribution of work hours is fixed, as

[9] This conceptual problem should be kept distinct from the empirical problem of estimating morbidity when workers are covered by paid sick leave. What is usually observed in morbidity studies is not sickness itself but days of illness reported. Sick leave obviously can affect whether an individual responds to illness by staying home from work.

in the 9-to-5 workday. In the previous model illness affects utility directly, by taking up time that could be devoted to labor or leisure, and by inducing out-of-pocket costs. With paid sick leave, the illness that occurs during working hours affects utility and medical costs but the individual's money income is not affected. If we assume that time spent ill is as likely to occur during working as during leisure hours, we can put the proportion q of illness occurring during leisure at $q = (T - W)/T$, the proportion of nonwork hours.

In this model the maximum number of hours that can be worked per year is W. The actual time spent working is $W - (1 - q)S$, the nominal work time less sick leave taken. The individual is no longer able to trade work for leisure without restriction, and the marginal value of leisure is no longer equal to the wage rate. The problem now is to maximize utility subject to both a money and a time constraint. The Lagrangian is

$$\mathscr{L} = U(X, L, S) + \lambda(Y - D - X - mS) + \gamma(T - W - L - qS). \quad (30)$$

The marginal change in income Y necessary to keep utility constant while pollution changes is again

$$\frac{dY}{dP} = -\frac{V_P}{V_Y} = -\frac{S_P}{S_D}. \quad (31)$$

In this case the employer also benefits from improvements in air quality, due to a reduction in sick-leave payments. Assuming that the wage rate equals the value of a day's output by a worker,[10] we can represent the benefit to the employer of a marginal reduction in pollution as

$$(1 - q)w\frac{dS}{dP}. \quad (32)$$

Combining (31) and (32), the social benefit of a reduction in pollution, for each worker in the population, is

$$\text{benefit} = (1 - q)w\frac{dS}{dP} - \frac{S_P}{S_D}. \quad (33)$$

Above, we showed that the costs of illness plus reductions in defensive measures underestimate the true benefits of pollution control. Once sick leave is introduced, this may no longer be the case. From the first-order conditions for maximizing (30), we have

$$m = \frac{U_S}{\lambda} - \frac{1}{S_D} - \frac{\gamma}{\lambda}q. \quad (34)$$

[10] If the employer expects a certain number of sick days each year and sets wages accordingly, the value of a day's output might exceed the wage rate slightly.

The benefit estimate given by the cost-of-illness approach is

$$\begin{aligned}\mathrm{COI} &= [w + m]\frac{dS}{dP} \\ &= w\frac{dS}{dP} + \frac{U_S}{\lambda}\frac{dS}{dP} - \frac{\gamma}{\lambda}q\frac{dS}{dP} - \frac{1}{S_D}[S_D D_P + S_P]. \end{aligned} \tag{35}$$

Total benefits B equal the sum of those occurring to the worker and his employer, and in this case equal

$$B = \text{cost of illness} + D_P - \frac{U_S}{\lambda}\frac{dS}{dP} + \left(\frac{\gamma}{\lambda} - w\right)q\frac{dS}{dP}. \tag{36}$$

Therefore, the benefit associated with a change in P equals the cost of illness plus three additional terms. The first two, representing the change in defensive expenditures and the direct disutility of illness, have been encountered before. The third term is a correction that takes account of the fact that sickness outside of working hours is no longer valued at the wage rate (the ratio γ/λ is the individual's marginal value of leisure time). If leisure has a value exceeding the wage rate, this term is positive and we can again conclude that the cost-of-illness plus defensive expenditures underestimate benefits. Even if leisure time is given a zero value, defensive expenditures plus the cost of illness adjusted by the time that the individual actually spends at work provides an underestimate. In fact, most COI studies implicitly give leisure a zero value by counting only the duration of missed work or by valuing a day of illness at the daily wage, even though individuals work only part of the day.

4. CONCLUSION

Our principal objective in this paper has been to compare ad hoc approaches to health benefit estimation with an expression of willingness to pay derived from a model of individual utility maximization. The most prominent of the ad hoc approaches is what we have called the cost of illness, which sums out-of-pocket health care expenses and lost wages. Another is the notion of defensive expenditures, discussed directly by Fisher and Zeckhauser [5], Courant and Porter [3] and, in another context, Shibata and Winrich [12]. We have shown that the cost-of-illness plus defensive expenditures normally underestimate "true" benefits. We found it was possible for defensive expenditures alone to overestimate the willingness to pay to avoid increased pollution, but only if such expenditures improve health in spite of the pollution increase.

Our finding proved robust with respect to several variations in our basic model. True benefits exceed those derived from summing defensive expenditures and the cost of illness when wages were assumed to be affected by pollution-induced illness, and when pollution was assumed to enter the utility function both directly as well as indirectly via its effect on sickness and, hence, the labor–leisure tradeoff. Only when our basic model was extended to include the case of paid sick leave was there indeterminacy about the relationship between the traditional cost-of-illness plus defensive expenditures and willingness to pay as derived from the model. Even in this case we found an underestimate when valuation of lost wages was limited, as is customary, to time actually spent on the job.

A final observation concerns the estimability of the conceptually correct benefit measure. The expression that we derived for willingness to pay, $-S_P/S_D$, is observable in principle. Bringing this term to life, so to speak, is more difficult than may at first appear. It requires detailed information on both the market and the nonmarket measures that individuals take to defend themselves against pollution and other environmental threats. Until such data are available, our conclusions suggest that the cost-of-illness approach can be used as a lower bound for true benefits in most interesting cases.

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